

BAYESIAN MULTIVARIATE SPARSE FUNCTIONAL PRINCIPAL COMPONENTS ANALYSIS WITH APPLICATION TO LONGITUDINAL MICROBIOME MULTIOMICS DATA

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Microbiome researchers often need to model the temporal dynamics of multiple complex, nonlinear outcome trajectories simultaneously. This motivates our development of *multivariate Sparse Functional Principal Components Analysis* (mSFPCA), extending existing SFPCA methods to simultaneously characterize multiple temporal trajectories and their interrelationships. As with existing SFPCA methods, the mSFPCA algorithm characterizes each trajectory as a smooth mean plus a weighted combination of the smooth major modes of variation about the mean, where the weights are given by the component scores for each subject. Unlike existing SFPCA methods, the mSFPCA algorithm allows estimation of multiple trajectories simultaneously, such that the component scores, which are constrained to be independent within a particular outcome for identifiability, may be arbitrarily correlated with component scores for other outcomes. A Cholesky decomposition is used to estimate the component score covariance matrix efficiently and guarantee positive semidefiniteness given these constraints. Mutual information is used to assess the strength of marginal and conditional temporal associations across outcome trajectories. Importantly, we implement mSFPCA as a Bayesian algorithm using R and stan, enabling easy use of packages such as PSIS-LOO for model selection and graphical posterior predictive checks to assess the validity of mSFPCA models. Although we focus on application of mSFPCA to microbiome data in this paper, the mSFPCA model is of general utility and can be used in a wide range of real-world applications.

1. Introduction. Numerous disorders, including heritable immune-mediated diseases such as inflammatory bowel disease (IBD) and asthma, neurological conditions including autism, and genetically driven diseases such as cancer, have been linked to dysregulation of human microbiota (Holleran et al. (2018), Lloyd-Price et al. (2019), Frati et al. (2019), Sharon et al. (2019), Ballen et al. (2016)). However, the complex influences of microbiota on human health are not yet functionally understood. To understand the links between the human microbiome and disease, it is necessary to determine which microbe genes are being expressed and the timing of their expression (Sberro et al. (2019)). Thus, in addition to obtaining microbiome data using 16S ribosomal RNA gene sequencing or whole genome shotgun sequencing (Kuczynski et al. (2010), Ranjan et al. (2016), Gill et al. (2006)), an increasing number of studies are also collecting transcriptomics data to understand microbial gene expression, proteomics data to study expressed proteins, and metabolomics data to define

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the functional status of host-microbial relationships (Integrative HMP (iHMP) Research Network Consortium (2014), Lloyd-Price et al. (2019), Bouslimani et al. (2019)). This complex combination of data types, called *microbiome multiomics*, is essential for understanding the links between microbial communities and disease and may enable translation of microbiome research into effective treatments.

An increasing number of microbiome multiomics studies are longitudinal, aimed at simultaneously characterizing microbiome and host temporal changes to provide a more comprehensive picture of dynamics during healthy and diseased states (Integrative HMP (iHMP) Research Network Consortium (2014), Lloyd-Price et al. (2019), Vatanen et al. (2018), Stewart et al. (2018)). Despite these breakthroughs in microbiome study designs and data collection, few statistical methods are available to analyze these complex longitudinal omics data. Recently, several new methods based on network analysis were developed for multiomics integration of microbiome data in cross-sectional studies (Jiang et al. (2019), Morton et al. (2019)), however, analytical methods for longitudinal microbiome multiomics data are still in their infancy. The challenges include irregular timing and frequency across subjects, unmatched time points between different data types, nonlinear temporal patterns, missing data and high individual variability (Bodein et al. (2019)).

The statistical framework of functional data analysis (FDA) was introduced by Ramsay and Silverman (Ramsay and Silverman (1997)), wherein the basic unit of information is the entire function, such as a curve or an image. Functional principal component analysis (FPCA) has been a widely used tool in FDA. The fundamental aim of FPCA is to reduce dimensionality by capturing the principal modes of smooth variation. FPCA summarizes subject-specific deviations from the mean curve via the coordinates (*principal component scores*) of the basis spanned by the principal components (Di et al. (2009)). Existing FPCA methods include smoothed FPCA approaches based on roughness penalties (Rice and Silverman (1991)), extensions to sparsely sampled functional data (James, Hastie and Sugar (2000), Yao, Müller and Wang (2005), Peng and Paul (2009), Di, Crainiceanu and Jank (2014), Kidziński and Hastie (2018)), and the asymptotic properties of FPCA (Hall and Hosseini-Nasab (2006), Li and Hsing (2010)). Most FPCA methods development has focused on univariate functional data. Chiou, Chen and Yang (2014) proposed a multivariate FPCA method to simultaneously model multiple temporal outcomes and infer the component dependencies through pairwise cross-covariance functions. However, this method is limited to functional data as classically conceived, where the curves are observed longitudinally over densely and consistently sampled time points.

To meet the need for modeling irregularly and sparsely sampled, nonlinear multivariate microbiome multiomics trajectories, we developed multivariate sparse functional principal components analysis (mSFPCA). The major novelty of our approach is that it focuses on a set of functions, which may covary in complex ways. Smoothing is accomplished by retaining a low-dimensional set of PC functions, as in James, Hastie and Sugar (2000). The PC scores across outcomes are modeled jointly via a constrained covariance matrix, efficiently estimated by Cholesky decomposition. Our proposed method allows for simultaneously characterizing multiple data modalities, such as microbiome, metabolome, inflammatory markers, and self-report measures, and to infer the temporal associations among these measures both marginally and conditionally, based on estimation of marginal and partial mutual information. The model is implemented using a Bayesian formulation, instantiated with Hamiltonian Markov Chain Monte Carlo (MCMC) methods in stan to sample from the posterior distribution of the model parameters. Our Bayesian implementation enables the usage of PSIS-LOO for model selection and graphical posterior predictive checks to assess the validity of mSFPCA models. While we focus on application of mSFPCA to microbiome data in this paper, the mSFPCA model is of general utility and can be used in many applications.

The remainder of the paper is organized as follows. Section 2 reviews sparse functional principal component analysis (SFPCA), and introduces multivariate SFPCA (mSFPCA), our statistical framework for longitudinal microbiome multiomics data. Section 3 describes extensive simulation studies to evaluate the performance of mSFPCA in realistic settings. Section 4 describes the application of our methodology to a challenging longitudinal microbiome multiomics data on type 2 diabetes. Section 5 presents our conclusion. To ensure reproducibility of our results, accompanying software, simulations and analysis results are provided in the Supplementary Material (Jiang et al. (2022)), as well at <https://github.com/knightlab-analyses/mfPCA-analyses>.

2. Methodology.

2.1. Sparse functional principal components analysis. The classical assumption of functional data analysis is that each trajectory is sampled over a dense grid of time points common to all individuals (Ramsay and Silverman (2007)). However, in practice, trajectories are often measured at an irregular and sparse set of time points that can differ widely across individuals. To address this issue, James, Hastie and Sugar (2000) proposed *sparse functional principal components analysis* (SFPCA) using a reduced rank mixed-effects framework. Let $Y_i(t)$ be the measurement at time t for the i th individual, $\mu(t)$ the overall mean function, f_j the j th principal component function and $f = [(f_1, f_2, \dots, f_k)]^T$, where k is the number of principal components. Then the James, Hastie and Sugar (2000) SFPCA model is given by

$$(1) \quad Y_i(t) = \mu(t) + \sum_{j=1}^k f_j(t)\alpha_{ij} + \epsilon_i(t), \quad i = 1, \dots, N$$

subject to the orthogonality constraint $\int f_j f_l = \delta_{jl}$, the Kronecker δ . The vector $\alpha_i = (\alpha_{i1}, \dots, \alpha_{ik})^T$ is the component weights for the i th individual and $\epsilon_i(t)$ is a normally-distributed residual, independent across subjects and across times within subject. The functions μ and f are approximated using cubic splines to allow a smooth but flexible fit. Let $b(t)$ be a cubic spline basis with dimension $q > k$. The spline basis is orthonormalized so that $\int b_j b_l = \delta_{jl}$. Let Θ and θ_μ be, respectively, a $q \times k$ matrix and a q -dimensional vector of real-valued coefficients. For each individual i , denote their measurement times by $t = (t_{i1}, t_{i2}, \dots, t_{in_i})^T$, and let $Y_i = (Y_i(t_{i1}), \dots, Y_i(t_{in_i}))^T$ be the corresponding real-valued observations. Then $B_i = (b(t_{i1}), \dots, b(t_{in_i}))^T$ is the $n_i \times q$ spline basis matrix for the i th individual. The reduced rank model can then be written as

$$(2) \quad \begin{aligned} Y_i &= B_i \theta_\mu + B_i \Theta \alpha_i + \epsilon_i, \quad i = 1, \dots, N, \\ \Theta^T \Theta &= I, \quad \alpha_i \sim N(0, D), \quad \epsilon_i \sim N(0, \sigma_\epsilon^2 I_{n_i}), \end{aligned}$$

where the covariance matrix D is restricted to be diagonal and I_{n_i} is the $n_i \times n_i$ identity matrix. Various fitting approaches, such as the EM algorithm, kernel smoothing and Newton–Raphson algorithm, have been proposed to estimate parameters of the SFPCA model (James, Hastie and Sugar (2000), Yao, Müller and Wang (2005), Peng and Paul (2009)). These approaches then use model selection techniques, such as cross-validation, Akaike information criterion (AIC) and leave-one-curve-out cross-validation, to select the dimension of the basis and the number of principal components. However, due to their reliance on assumptions such as normally-distributed component scores and residuals, these models need to be carefully examined when applied to real data (Kidziński and Hastie (2018)).

2.2. Bayesian SFPCA. Jiang et al. (2020) proposed an SFPCA model in a Bayesian framework to allow for flexible prior specification and implementation of model selection and assessment methods. The real-valued observations $Y_i(t)$ are first standardized to have mean zero and standard deviation one. The prior distributions for parameters in equation (2) were chosen as follows:

$$\begin{aligned}\theta_\mu &\sim N_q(0, I_q), \\ \alpha_i &\sim N_k(0, I_k), \\ \Theta_j &\sim N_q(0, I_q), \quad j = 1, \dots, k, \\ \epsilon_i &\sim N_{v_i}(0, \sigma_\epsilon^2 I_{v_i}), \\ \sigma_\epsilon &\sim \text{Cauchy}(0, 1),\end{aligned}$$

where Θ_j is the j th column of the loading matrix Θ , and v_i is the total number of visits for the i th subject (note that v_i is different from the earlier notation n_i , with n_i indexing the last available time points for subject i). This Bayesian implementation enables use of leave-one-out cross-validation with Pareto-smoothed important sampling (PSIS-LOO) (Vehtari, Gelman and Gabry (2017)) to perform model selection on the number of principal components k and the number of basis functions q . This Bayesian implementation used the Hamiltonian MCMC sampling algorithm in Stan (Carpenter et al. (2017)). Model fit can be assessed via diagnostic plots from PSIS-LOO as well as the graphical posterior predictive checks obtained from simulating posterior predictive data (Gelman, Meng and Stern (1996), Gabry et al. (2019)). This Bayesian implementation thus offers a flexible and comprehensive solution to real-data SFPCA applications.

2.3. Multivariate SFPCA. Here, we directly extend our previous Bayesian SFPCA from modeling one-dimensional response of N subjects over time to simultaneously model multiple dimensional longitudinal responses from subjects using multivariate Functional PCA (mSFPCA), and to quantify their marginal and conditional temporal associations. Let $\mathbf{Y}_i(t)$ denote the P -dimensional observed response at time t for subject i , and let K_p be the number of FPCs for the p th response, $p = 1, \dots, P$. The total number of FPCs across the p responses are thus $K = \sum_{p=1}^P K_p$. We then model the multivariate responses as

$$(3) \quad \mathbf{Y}_i(t) = \boldsymbol{\mu}_i(t) + \mathbf{f}(t)^T \boldsymbol{\alpha}_i + \boldsymbol{\epsilon}_i(t), \quad i = 1, \dots, N,$$

where $\boldsymbol{\mu}_i = (\mu_{i,1}(t), \dots, \mu_{i,P}(t))^T$ is the overall mean response for subject i , $\mathbf{f}(t)^T = \text{diag}(f_1(t)^T, \dots, f_P(t)^T)$ is a $P \times K$ matrix of FPCs with $f_p(t)$ the K_p -dimensional vector of FPC functions corresponding to the p th response, $\boldsymbol{\alpha}_i = (\alpha_{i,1}^T, \dots, \alpha_{i,P}^T)^T$ is the K -dimensional vector of FPC scores for subject i , and $\boldsymbol{\epsilon}_i$ is the P -dimensional vector of corresponding residuals.

In order to fit this model when the data are sampled a finite number of time points, we use cubic spline basis functions to approximate $\boldsymbol{\mu}_i$ and $\mathbf{f}(t)$. Let Q_p be the number of basis functions, and let V_{ip} be the total number of assessments for i th subject in the p th trajectory, with assessment times $\mathbf{t}_{ip} = (t_{ip1}, \dots, t_{ipV_{ip}})^T$. Let $V_i = \sum_{p=1}^P V_{ip}$ be the total number of assessment times across all P responses for the i th subject. The multivariate Sparse Functional PCA (mSFPCA) model is then given by

$$(4) \quad \mathbf{Y}_i = \mathbf{B}_i \boldsymbol{\theta}_\mu + \mathbf{B}_i \boldsymbol{\Theta} \boldsymbol{\alpha}_i + \boldsymbol{\epsilon}_i, \quad i = 1, \dots, N,$$

where $\mathbf{Y}_i$ is a V_i -dimensional vector of observed responses for the i th subject, residuals $\boldsymbol{\epsilon}_i \sim N_{V_i}(0, \sigma_{\epsilon}^2 I_{V_i})$, $\mathbf{B}_i$ is a $V_i \times Q$ matrix with

$$\mathbf{B}_i = \begin{bmatrix} B_{i1} & 0^T & \cdots & 0^T \\ 0 & B_{i2} & \cdots & 0^T \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & B_{iP} \end{bmatrix},$$

where B_{ip} is the $V_{ip} \times Q_p$ matrix of basis functions evaluated at times $\mathbf{t}_{ip}$. The $Q \times K$ matrix $\boldsymbol{\Theta}$ of FPC loadings

$$\boldsymbol{\Theta} = \begin{bmatrix} \Theta_1 & 0^T & \cdots & 0^T \\ 0 & \Theta_2 & \cdots & 0^T \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \Theta_P \end{bmatrix},$$

is subject to the orthonormality constraint $\boldsymbol{\Theta}^T \boldsymbol{\Theta} = \mathbf{I}$. The K -dimensional vector of FPC scores $\boldsymbol{\alpha}_i \sim N(0, \Sigma_{\alpha})$, with Σ_{α} is restricted to the form

$$\begin{bmatrix} D_1 & C_{21}^T & \cdots & C_{P1}^T \\ C_{21} & D_2 & \cdots & C_{P2}^T \\ \vdots & \vdots & \ddots & \vdots \\ C_{P1} & C_{P2} & \cdots & D_P \end{bmatrix},$$

where D_p is the within-trajectory diagonal covariance matrix (necessary for identifiability of within trajectory FPCs) for the p th trajectory, and C_{lm} is the covariance matrix of the FPCs for the l th and m th responses. Σ_{α} can be written as $\Sigma_{\alpha} = S_{\alpha} R_{\alpha} S_{\alpha}$, where S_{α} is the diagonal matrix of standard deviations for the FPC scores, and R_{α} is the correlation matrix restricted to the form

$$\begin{bmatrix} I_1 & R_{21}^T & \cdots & R_{P1}^T \\ R_{21} & I_2 & \cdots & R_{P2}^T \\ \vdots & \vdots & \ddots & \vdots \\ R_{P1} & R_{P2} & \cdots & I_P \end{bmatrix},$$

where I_p is the $Q_p \times Q_p$ identity matrix.

We implemented the mSFPCA model using the Hamiltonian Markov Chain Monte Carlo (MCMC) sampling algorithm in Stan (Carpenter et al. (2017)) to estimate parameters. The prior distributions for $\boldsymbol{\theta}_{\mu}$, $\boldsymbol{\Theta}$ and $\boldsymbol{\epsilon}_i$ are set as follows:

$$\begin{aligned} \boldsymbol{\theta}_{\mu} &\sim N_Q(0, I_Q), \\ \boldsymbol{\Theta}_{kp} &\sim N_{Q_p}(0, I_{Q_p}), \\ \boldsymbol{\epsilon}_i &\sim N_{V_i}(0, \sigma_{\epsilon}^2 I_{V_i}), \\ \sigma_{\epsilon} &\sim \text{Cauchy}(0, 1), \end{aligned}$$

where $\boldsymbol{\Theta}_{kp}$ is the k th column of the FPC loadings in the p th block. Leveraging this Bayesian implementation in Stan, we utilized PSIS-LOO for model selection, and diagnostics plots from PSIS-LOO and graphical posterior predictive checks for model diagnostics.

2.3.1. *Orthonormality constraints.* A difficulty in implementing the Bayesian mSFPCA model is that the principal component loadings Θ are not uniquely specified. For a given $K \times K$ rotation matrix P , if $\Theta^* = \Theta P$ and Θ obeys the constraints in equation (4), then $\Theta^{*T} \Theta^* = P^T \Theta^T \Theta P = I$, and hence Θ is unidentifiable without additional restrictions. Instead of directly enforcing orthonormality when sampling from the conditional posteriors in the Bayesian model fitting, we sampled the parameters with no constraint on Θ and then performed a *post hoc* rotation for each iteration of the MCMC algorithm to meet the orthonormality constraint. Since the symmetric matrix $\Theta \Sigma_\alpha \Theta^T$ is identifiable and nonnegative definite, we applied an eigenvalue decomposition $\Theta \Sigma_\alpha \Theta^T = V S V^T$, where V is the $Q \times Q$ matrix of orthonormal eigenvectors, and S is the diagonal matrix of eigenvalues, with the Q positive eigenvalues ordered from largest to smallest. Let $\Theta^* = V_k$ denote the $Q \times K$ matrix consisting of the first K eigenvectors of V , which satisfies $\Theta^{*T} \Theta^* = I$. Finally, we rotated Σ_α and FPC scores α_i , to obtain $\Sigma_\alpha^* = \Theta^{*T} \Theta \Sigma_\alpha \Theta^T \Theta^*$ and $\alpha_i^* = \Theta^{*T} \Theta \alpha_i$, so that $\Theta^* \Sigma_\alpha^* \Theta^{*T} = \Theta \Sigma_\alpha \Theta^T$ and $\Theta^* \alpha_i^* = \Theta \alpha_i$.

2.3.2. *Modeling covariance.* Since the covariance matrix of FPC scores Σ_α must be positive semidefinite and is restricted to be diagonal within each response, it is a challenge to model this covariance matrix effectively. Barnard, McCulloch and Meng (2000) proposed a separation strategy for modeling $\Sigma = SRS$ by assuming independent priors for the standard deviations S and the correlation matrix R . To account for the dependent structure of correlations among different subsets of variables, Liechty, Liechty and Müller (2004) proposed the common correlation model for R , which assumes a common normal prior for all correlations with the additional restriction that the correlation matrix is positive definite. However, the awkward manner in which r_{ij} , the ij th element in the correlation matrix R , is embedded in the full conditional posterior density, leads to use of a Metropolis–Hastings algorithm to update one coefficient r_{ij} at a time (Liechty, Liechty and Müller (2004)). This consecutive updating procedure for correlation estimation is inefficient, and could lead to heavy computational cost when the correlation matrix is large or when the correlation has to be estimated separately from other parameters in mSFPCA model when implemented in Stan (Carpenter et al. (2017)). For example, in a simulated data with 3 data modalities from 100 subjects over 10 time points, it would take 40 hours for a mSFPCA model using Liechty’s covariance estimation method to estimate all the parameters when implemented in Stan. However, the computational time can be reduced over 130 times (to only 18 minutes) by using our proposed method due to the avoidance of additional Metropolis–Hastings algorithm. Our proposed algorithm utilizes the Cholesky decomposition of the constrained FPC covariance matrix. Details of the sampling algorithm are given in the Supplementary Material.

2.3.3. *Estimating interblock association.* Apart from simultaneously modeling multivariate longitudinal measurements, we want to estimate the association among measurements of interest via the correlations among the FPC scores, where the correlation matrix R_α obtained earlier will play a crucial role. We propose a measure of intertrajectory association by calculating the mutual information of FPC scores from different measurements.

We define the intertrajectory association between measurements p_1 and p_2 as the mutual information of FPC scores α_{ip_1} and α_{ip_2} , $1 \leq p_1, p_2 \leq P$, with

$$MI(\alpha_{ip_1}, \alpha_{ip_2}) = H(\alpha_{ip_1}) + H(\alpha_{ip_2}) - H(\alpha_{ip_1}, \alpha_{ip_2}),$$

where $H(X)$ is the entropy of X and $H(X) = -E[\log(f_X(X))]$ with $f_X(X)$ being the probability density function of X (Cover (1999)).

If K -dimensional random variable X follows multivariate normal distribution with covariance matrix Σ , then according to [Ahmed and Gokhale \(1989\)](#),

$$H(X) = \frac{k}{2} + \frac{k}{2} \log(2\pi) + \frac{1}{2} |\Sigma|.$$

Since the K -dimensional FPC scores $\alpha_i \sim N(0, \Sigma_\alpha)$, and any subvector of α_i is of the same structure with the correlation matrix being a submatrix of R_α , then according to [Arellano-Valle, Contreras-Reyes and Genton \(2013\)](#), the mutual information of α_{ip_1} and α_{ip_2} could be simplified as

$$(5) \quad MI(\alpha_{ip_1}, \alpha_{ip_2}) = -\frac{1}{2} \log |R_{\alpha\{p_1, p_2\}}|,$$

where

$$R_{\alpha\{p_1, p_2\}} = \begin{vmatrix} I_{p_1} & R_{p_1 p_2} \\ R_{p_1 p_2}^T & I_{p_2} \end{vmatrix}.$$

Moreover, we can estimate the conditional intertrajectory association between any two measurements of interest given the other measurements in the model by calculating the partial mutual information of FPC scores. The conditional intertrajectory association between measurements p_1 and p_2 is defined as the partial mutual information of α_{ip_1} and α_{ip_2} , $1 \leq p_1, p_2 \leq P$, with

$$\begin{aligned} & MI(\alpha_{ip_1}, \alpha_{ip_2} | \alpha_{i\{1, \dots, P \setminus p_1, p_2\}}) \\ &= H(\alpha_{ip_1}, \alpha_{i\{1, \dots, P \setminus p_1, p_2\}}) + H(\alpha_{ip_2}, \alpha_{i\{1, \dots, P \setminus p_1, p_2\}}) \\ & \quad - H(\alpha_{i\{1, \dots, P \setminus p_1, p_2\}}) - H(\alpha_{ip_1}, \alpha_{ip_2}, \alpha_{i\{1, \dots, P \setminus p_1, p_2\}}) \\ (6) \quad &= H(\alpha_{i\{1, \dots, P \setminus p_2\}}) + H(\alpha_{i\{1, \dots, P \setminus p_1\}}) - H(\alpha_{i\{1, \dots, P \setminus p_1, p_2\}}) - H(\alpha_i) \\ &= \frac{1}{2} \log |R_{\alpha\{1, \dots, P \setminus p_2\}}| + \frac{1}{2} \log |R_{\alpha\{1, \dots, P \setminus p_1\}}| \\ & \quad - \frac{1}{2} \log |R_{\alpha\{1, \dots, P \setminus p_1, p_2\}}| - \frac{1}{2} \log |R_\alpha|, \end{aligned}$$

where $R_{\alpha\{1, \dots, P \setminus p_2\}}$, $R_{\alpha\{1, \dots, P \setminus p_1\}}$, and $R_{\alpha\{1, \dots, P \setminus p_1, p_2\}}$ are defined in the similar way as $R_{\alpha\{p_1, p_2\}}$ in equation (5).

Intertrajectory association obtained from this way ranges from 0 to infinity. By analogy with the way Person's contingency coefficient was obtained, we can apply a simple transformation proposed by [Joe \(1989\)](#) to obtain a normalized version of the mutual information as

$$(7) \quad MI^*(\alpha_{ip_1}, \alpha_{ip_2}) := \sqrt{1 - \exp[-2MI(\alpha_{ip_1}, \alpha_{ip_2})]}.$$

In this way, the intertrajectory and conditional associations now take its value in $[0, 1]$. The interpretation is that the closer $MI^*(\alpha_{ip_1}, \alpha_{ip_2})$ or $MI^*(\alpha_{ip_1}, \alpha_{ip_2} | \alpha_{i\{1, \dots, P \setminus p_1, p_2\}})$ is to 1, the higher the temporal association between measurements is.

3. Simulation studies. To evaluate the performance of mSFPCA in modeling multiple data modalities, especially in its covariance estimation and temporal association inference, we simulated sparse longitudinal trajectories with three data modalities under four different covariance structures. To better mimic the reality, our data was simulated based on an mSFPCA model using parameters initially estimated from a real longitudinal microbiome multiomics dataset ([Kostic et al. \(2015\)](#)) in the following way:

1. Applying mSFPCA to model three data modalities in the real multiomics data set.
2. Selecting the optimal number of PCs and dimension of basis using PSIS-LOO: the chosen model has the number of PCs as 2, 2, 1, and the number of basis as 6, 5, 5 for each measurement, respectively.
3. Extracting the estimated values for population mean curve (θ_μ), FPC loadings (Θ) and residual variance σ_ϵ .

Then under four distinct covariance structures on FPC scores (Σ_α), we simulate the trajectories for 100 subjects with an average of 20% missing data over observations at 10 time points. Observations were randomly deleted to create increasingly sparse functional data sets. In the 1st covariance structure, all PCs are independent; in the 2nd covariance structure, only 1 strong correlation of 0.75 exists across all PCs; in the 3rd covariance structure, 1 strong and 1 medium correlation exists, at values of 0.75 and 0.5, respectively; in the 4th covariance structure, 1 strong, 1 medium and 1 weak correlations exists at strength of 0.75, 0.5 and 0.25. In short, there are increasing dependence structures among PCs as the covariance structure moves from the first to the last. Based on these prespecified covariance structures and initially estimated parameters, we simulate the sparse longitudinal trajectories as follows:

1. Choosing the total number of subjects to be 100, and the number of time points to be 10 in order to place possible time points between $[0, 1]$.
2. Simulating the observed number of time points for each individual with $n_i \sim \text{Poisson}(8)$, where 8 represents the average number of time points across all subjects, and then randomly placing the observed time points in the possible time locations (chosen in the previous step).
3. Generating the cubic spline basis matrix B_i for each subject (orthonormality obtained through Gram–Schmidt orthonormalization).
4. Simulating for each subject FPC scores $\alpha_i \sim N(0, \Sigma_\alpha)$ and noise $\epsilon_i \sim N(0, \sigma_\epsilon^2 I)$.
5. Obtaining the temporal trajectory for each individual with $Y_i = B_i \theta_\mu + B_i \Theta \alpha_i + \epsilon_i$.
6. Repeating step 1–5 1000 times for each covariance structure, thus generating 4000 simulated data sets in total.

To evaluate the mSFPCA model performance in simulated data, we want to examine three main results: (1) how well mSFPCA can capture the temporal patterns embodied in the overall mean curve and FPC curves for each temporal measurement; (2) the accuracy of covariance estimation; (3) the inference on temporal associations based on mutual information estimation.

Figure 1 shows that the estimated overall mean curves and PC curves accurately recovered the ground truth for all three outcome variables under covariance structure I. This accurate capturing of major temporal patterns was seen in other three covariance structures as well (Supplementary Figures 1–3). Figure 2 summarizes the performance of covariance estimation across all 4 scenarios in terms of the coverage probabilities of 95% credible intervals on estimated covariance parameters. The coverage probabilities are lowest in the 1st covariance structure (independent, Figure 2A), improved when more dependence structures are introduced (Figure 2B–D), and reach highest with the 4th covariance structure (having most correlations across PCs, Figure 2D). Despite these subtle differences in the coverage probabilities for each covariance parameter, the average coverage probability across all estimated parameters, represented by the dotted line, is around 95% within each covariance structure. This indicates that our mSFPCA model is able to estimate the covariance matrix properly, and its performance is affected by the structure of covariance matrix itself: the more sparse the covariance is, the more challenging the estimation. But even with the most sparse scenario (Figure 2A), our mSFPCA model is still able to achieve about 95% average coverage probability.

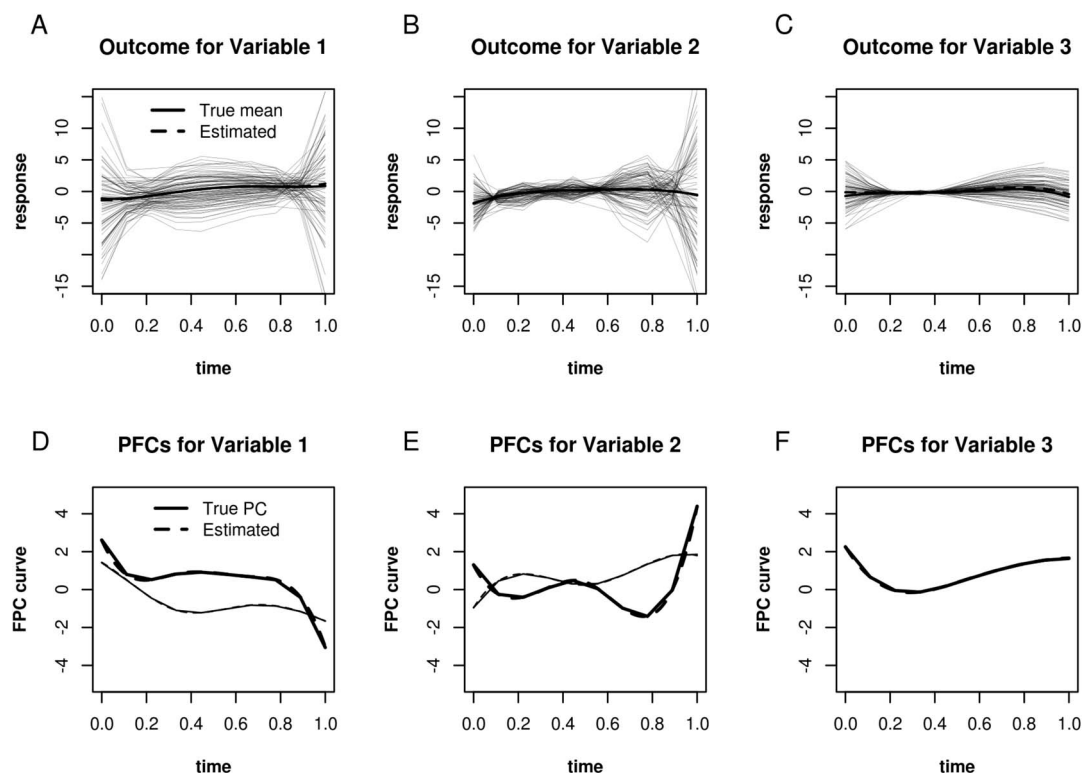


FIG. 1. Estimated mean and FPC curves from mSFPCA on simulated data with covariance structure I. Estimated (dashed) versus true (solid) overall mean curve on simulated trajectories (thin) on outcome variable 1 (A), outcome variable 2 (B), and outcome variable 1 (C). Estimated (dashed) versus true (solid) PC curves (PC 1 – thin line, PC 2 – thick line) on simulated data for outcome variable 1 (D), outcome variable 2 (E), and outcome variable 3 (F).

Regarding the inference on temporal associations, Table 1 shows the mutual information estimates in each simulation scenario, which estimates the temporal association between each pair of data modalities. MI_{ij} denotes the normalized mutual information between i th and j th data modalities. When the true MI is zero, the coverage probability is denoted as 0^* , as the estimated mutual information is always nonnegative, and thus by construction the posterior distribution will not encompass zero. In the rest of the table, except for the slightly lower coverage probability with 0.92 for MI_{12} in the 3rd scenario, or 0.93 for MI_{23} in the 4th scenario, all the coverage probabilities are close to 95%. Table 2 shows the conditional mutual information estimates in each simulation scenario, which estimates the temporal association between each pair of data modalities given the other measurement in the model. CMI_{ij} denotes the normalized conditional mutual information between i th and j th data modalities. All coverage probabilities of nonzero CMIs are close to 95% on the estimation of conditional mutual information.

In short, our simulation results have demonstrated the good performance of mSFPCA in modeling sparse longitudinal data with multiple data modalities and providing valid inference on temporal associations.

4. Real data application. For the real data application, we want to model multiple data modalities simultaneously in a large and challenging data set, with a special interest in utilizing conditional mutual information to infer temporal association. This data set comes from the type 2 diabetes (T2D) longitudinal studies in the Integrative Human Microbiome Project (Integrative HMP (iHMP) Research Network Consortium (2014)). In this example, an over

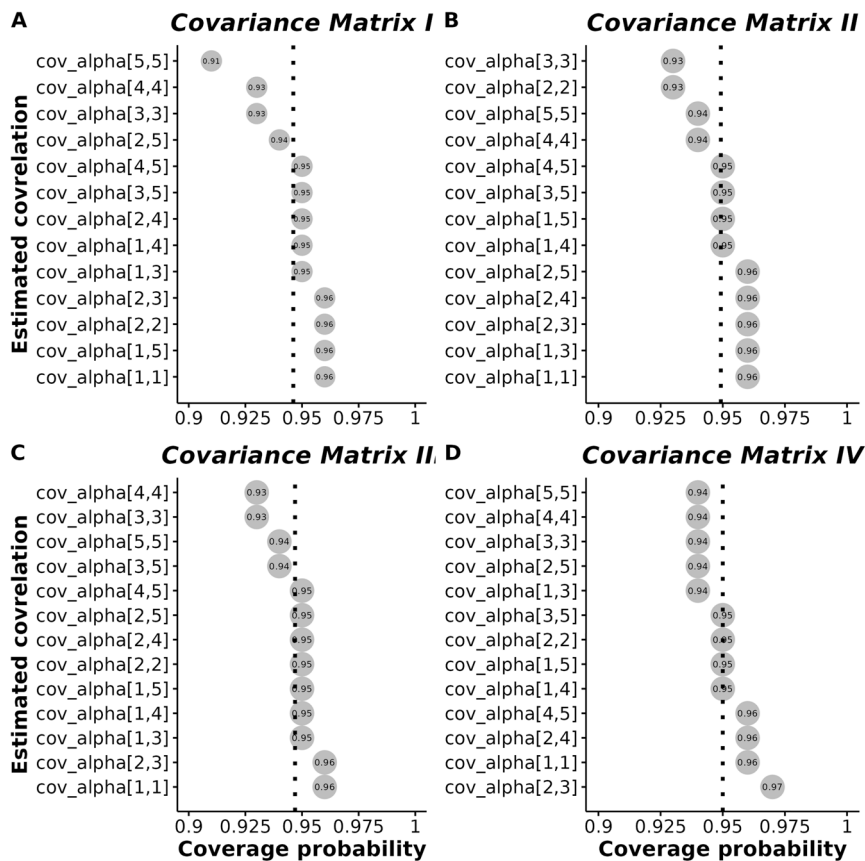


FIG. 2. Coverage probability of 95% credible interval on estimated covariance parameters in covariance matrix I(A), II(B), III(C) and IV(D). Values within each dot represent the coverage probability for each estimated covariance parameter, and only values for unique covariance element are displayed. Black dotted lines indicate the average coverage probability across all estimated parameters within each covariance matrix, which is around 95% in all four simulation scenarios.

TABLE 1
Mutual information estimates for each simulation scenario

Simulation scenario	Parameter	Truth	Median	95% credible interval		
				Cov. prob.	2.5%	97.5%
Covariance I	MI_{12}	0	0.26	0*	0.11	0.42
	MI_{13}	0	0.17	0*	0.04	0.34
	MI_{23}	0	0.18	0*	0.04	0.35
Covariance II	MI_{12}	0	0.26	0*	0.11	0.42
	MI_{13}	0	0.17	0*	0.04	0.34
	MI_{23}	0.75	0.75	0.96	0.65	0.83
Covariance III	MI_{12}	0.5	0.54	0.92	0.39	0.66
	MI_{13}	0	0.17	0*	0.04	0.34
	MI_{23}	0.75	0.75	0.96	0.65	0.83
Covariance IV	MI_{12}	0.5	0.54	0.94	0.38	0.66
	MI_{13}	0.25	0.29	0.94	0.12	0.46
	MI_{23}	0.75	0.75	0.93	0.66	0.83

TABLE 2
Conditional mutual information estimates for each simulation scenario

Simulation scenario	Parameter	Truth	Median	95% credible interval		
				Cov. prob.	2.5%	97.5%
Covariance I	CMI_{12}	0	0.26	0*	0.11	0.42
	CMI_{13}	0	0.17	0*	0.04	0.34
	CMI_{23}	0	0.18	0*	0.04	0.35
Covariance II	CMI_{12}	0	0.26	0*	0.12	0.42
	CMI_{13}	0	0.17	0*	0.04	0.34
	CMI_{23}	0.75	0.75	0.95	0.65	0.83
Covariance III	CMI_{12}	0.76	0.77	0.94	0.68	0.84
	CMI_{13}	0.66	0.66	0.95	0.53	0.76
	CMI_{23}	0.87	0.87	0.96	0.81	0.91
Covariance IV	CMI_{12}	0.81	0.82	0.95	0.74	0.88
	CMI_{13}	0.76	0.76	0.95	0.66	0.83
	CMI_{23}	0.89	0.90	0.95	0.85	0.93

3-year study has been conducted in approximately 100 individuals at high risk for T2D, in order to better understand the biological changes that occur during the onset and progression of T2D. Multiple sample types were collected from the study participants every 2–3 months during their healthy periods, with more frequent sampling during periods of respiratory illness and other environmental stressors. These data include multiomics assays such as stool microbiome data using 16S rRNA sequencing, host protein expression profiles in fecal samples using LC-MS/MS and cytokine profiles that quantify the levels of 50 diverse inflammatory proteins and insulin peptides in host serum, as well as standard clinical tests results like hemoglobin A1c (HbA1c), insulin and glucose. Moreover, behavior changes of patients, such as emotional and psychological stress, were documented using the Perceived Stress Scale instrument. Our outcomes of interest are the longitudinal pattern of Shannon diversities in bacteria, proteins and cytokines, and of clinical test results on HbA1c. Shannon diversity is defined as $Shannon = -\sum_{i=1}^S p_i \ln(p_i)$, where S is the total number of species, and p_i is the relative proportion of species i relative to the entire population. Shannon diversities in bacteria, proteins and cytokines are chosen over specific features in our application, because they have higher predictive power of individuals' diabetic status than specific bacteria, proteins or cytokines. Hence, we are particularly interested in utilizing mutual information to investigate which omics data (based on Shannon diversity) have strongest association with HbA1c, and whether additional omics data improve the temporal association based on conditional mutual information.

The estimated mean curves in Figure 3 show different temporal trends in each outcome, where Shannon bacterial diversity decreases slowly over time, protein diversity increases steadily over time, cytokine diversity increases over the first 300 days, decreases between day 300 and 900, and then increases, and HbA1c decreases during the first 2 years, and increases slightly afterwards. As indicated by the observed trajectories for each individual (black curves), there are great subject-level variations in each outcome. This additional temporal information is captured by the FPC curves in Figure 4. Figure 4A shows the first two PCs in Shannon bacterial diversity, of which PC 1 explains 79% and captures variation around day 750, while PC 2 explains 21% of the variation and emphasizes variation around day 300 and 1100. Figure 4B shows the first four PCs in Shannon protein diversity, of which the first two PCs explain over 90% of the variance. The first PC captures variation around day 750,

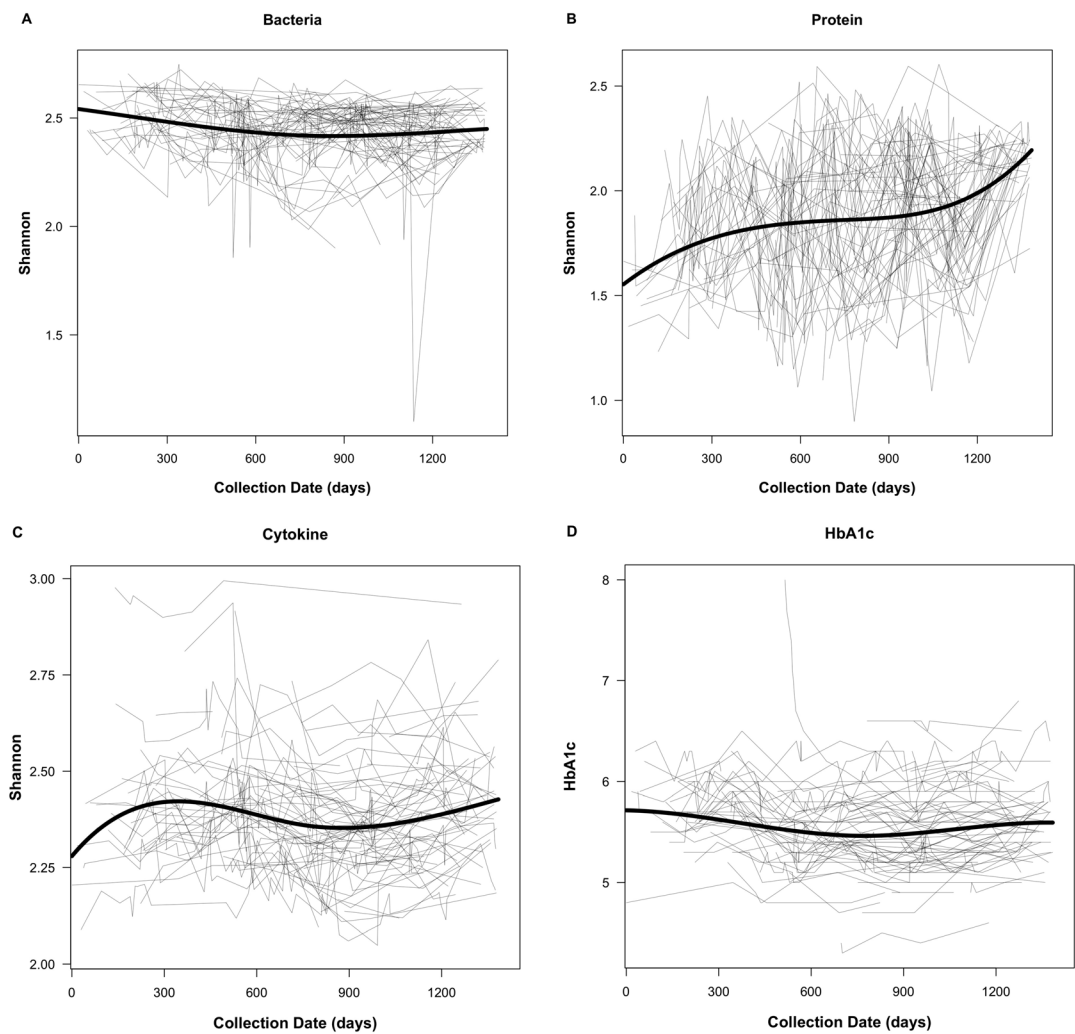


FIG. 3. Estimated mean curves from mSFPCA application on type 2 diabetes multiomics data set. (A) Estimated sample mean curve (dark) for Shannon bacterial diversity on observed individual trajectories (light). (B) Estimated sample mean curve (dark) for Shannon protein diversity on observed individual trajectories (light). (C) Estimated sample mean curve (dark) for Shannon cytokine diversity on observed individual trajectories (light). (D) Estimated sample mean curve (dark) for HbA1c on observed individual trajectories (light).

and the second PC emphasizes variation around day 450 and 1200. Figure 4C shows the first four PCs in Shannon cytokine diversity, of which PC 1 explains 83% and exhibits an almost flat curve over time, while PC 2 explains 13% of the variation and emphasizes variation around day 300 and 800. Figure 4D shows the first four PCs in HbA1c: PC1 exhibits a slight increasing curve over time, accounting for 70% variation, and PC 2 captures variation around day 300 and 800, explaining for an additional 21% variation. In short, although principal patterns in each measurement vary, changing time points are pretty consistent, suggesting coherent responses to changes in patients’ mental or physical conditions.

Among omics’ temporal associations with standard clinical test result HbA1c, Table 3 suggests that Shannon protein diversity has the highest association with HbA1c, at an estimated mutual information of 0.91 with 95% credible interval (0.786, 0.971). Cytokine diversity is the second highest, with MI at 0.849 (0.668, 0.954), and bacteria diversity has the lowest association at 0.71 (0.441, 0.854). However, when information about other omics measurements are provided, all the pairwise temporal associations increase to over 0.95, as indicated

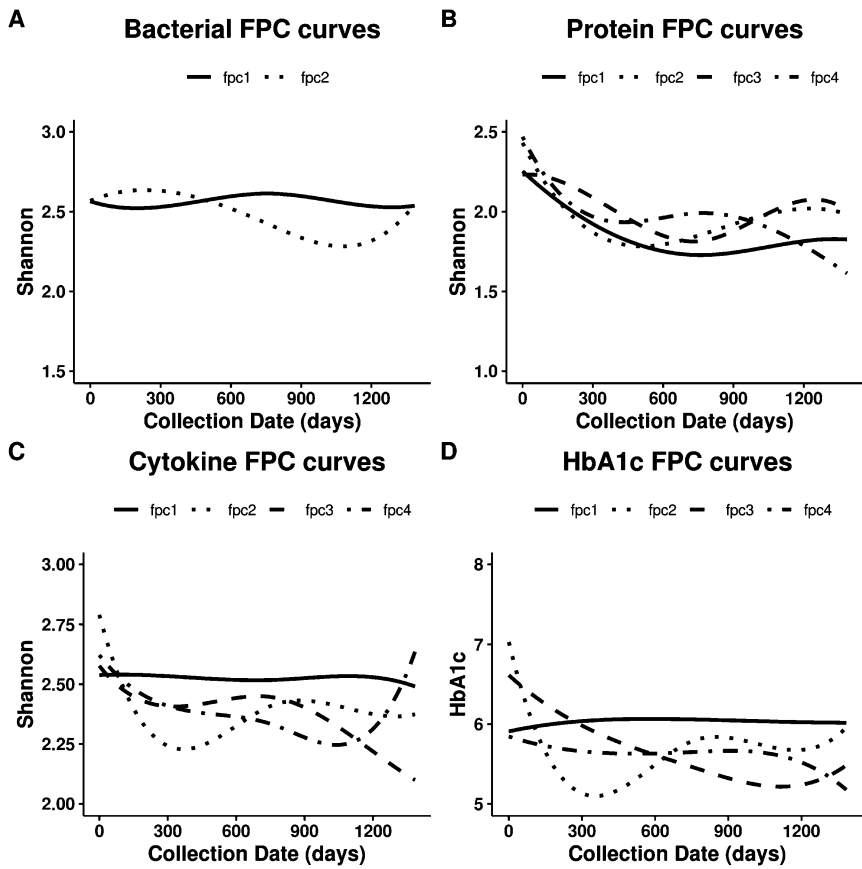


FIG. 4. Estimated FPC curves from mSFPCA application on type 2 diabetes multiomics data set. (A) Modes of variation in Shannon bacterial diversity captured by 2 principal component curves. (B) Modes of variation in Shannon protein diversity captured by 4 principal component curves. (C) Modes of variation in Shannon cytokine diversity captured by 4 principal component curves. (D) Modes of variation in HbA1c captured by 4 principal component curves.

by the conditional MI results. Regarding temporal associations among omics measurement, Shannon protein and bacteria diversities have highest temporal association, with mutual information at 0.982 (0.875, 0.999); Shannon protein and cytokine diversities also have high association at 0.966 (0.886, 0.998); the association between Shannon bacteria and cytokine is medium at 0.798 (0.454, 0.977). Similar to earlier results, when information about other

TABLE 3
Mutual information estimates for type 2 diabetes multiomics data set application

Temporal associations with HbA1c	MI (95% CI)	CMI (95% CI)
HbA1c—protein	0.910 (0.786, 0.971)	0.994 (0.968, 0.999)
HbA1c—cytokine	0.849 (0.668, 0.954)	0.986 (0.951, 0.999)
HbA1c—bacteria	0.710 (0.441, 0.854)	0.957 (0.820, 0.999)
Temporal associations among omics	MI (95% CI)	CMI (95% CI)
protein—bacteria	0.982 (0.875, 0.999)	0.999 (0.996, 0.999)
protein—cytokine	0.966 (0.886, 0.998)	0.999 (0.997, 0.999)
bacteria—cytokine	0.798 (0.454, 0.977)	0.995 (0.958, 0.999)

measurements are available, all conditional information increase to 0.99. In short, host protein expression profiles data has highest temporal association with patients’ diabetes status (i.e., HbA1c), but this information can still be further improved with additional omics data.

We need model diagnostics to conclude on the validity of our mSFPCA application. The optimal model selected by PSIS-LOO has 2 PCs for Shannon bacterial diversity, and 4 PCs for the other measurements, and the number of internal knots is chosen to be one for all outcomes. PSIS-LOO diagnostics in Figure 5A show that the selected mSFPCA model fit the majority of the data well, except for 4 outliers with Pareto shape k values higher than the warning threshold 0.7. Graphical posterior predictive checks in Figure 5B suggests good model fit as the simulated data from the posterior predictive distribution was able to cover the

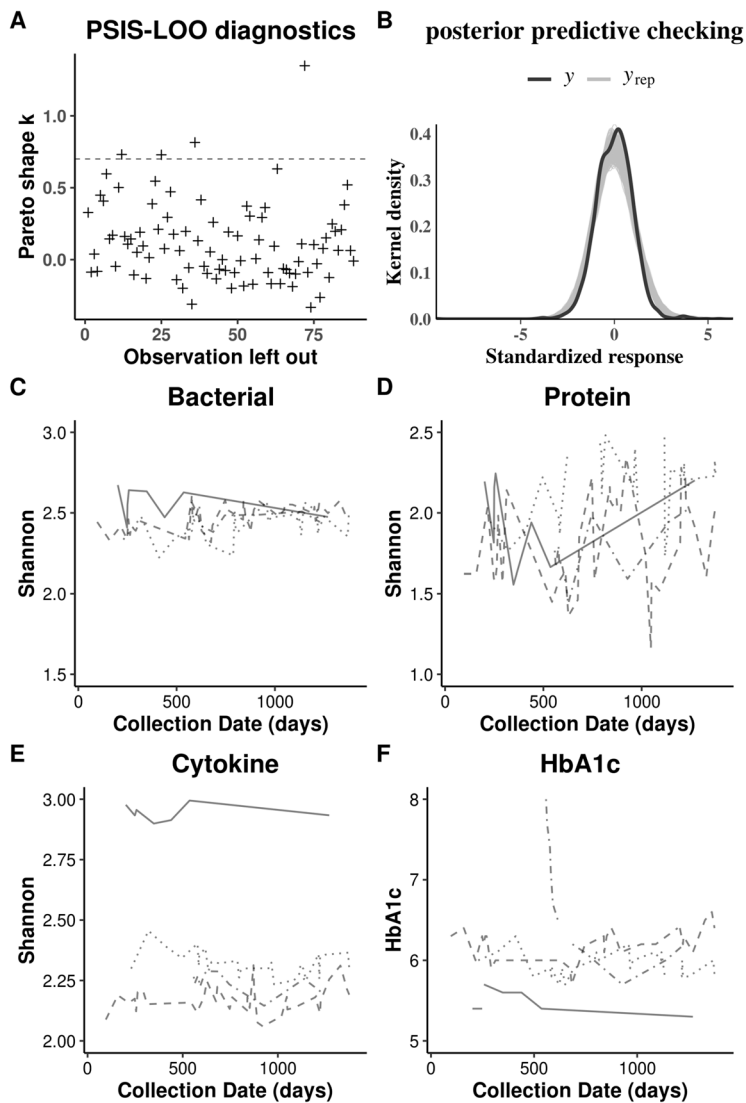


FIG. 5. Graphical model diagnostics and examination of outliers for mSFPCA application on type 2 diabetes multiomics data set. (A) Scatter plot of estimated Pareto shape parameter $\hat{k}$ in PSIS-LOO diagnostic plot: all $\hat{k}$'s except 4 are lower than the warning threshold 0.7. (B) Graphical posterior predictive plot: kernel density estimate of the observed data set y (dark curve), with kernel estimates for 100 simulated data set y_{rep} drawn from the posterior predictive distribution (thin, lighter lines). (C) Observed Shannon bacterial diversity for 4 outliers detected by PSIS-LOO diagnostic plot in (A). (D) Observed Shannon protein diversity for 4 outliers. (E) Observed Shannon cytokine diversity for 4 outliers. (F) Observed HbA1c for 4 outliers.

distribution of observed outcomes well. Figure 5C–F highlight the observed trajectories of the 4 outliers detected by PSIS-LOO diagnostic plot. The subject in solid line has the highest curve in Shannon cytokine diversity and low value in HbA1c. A closer look at his/her meta-data shows that this subject went through stages of healthy, infection and back to healthy. The subject in dotted line shows high oscillation pattern in Shannon protein diversity, as he/she oscillated between stages of healthy, inflammation and infection. The subject in dashed line exhibits high oscillation pattern in Shannon protein diversity, because he/she went through a complicated interweaving stages of healthy, inflammation, infection, post-travel and allergy. The subject in dot-dash line, who has the highest Pareto shape k value in Figure 5A, experienced drastic change in HbA1c, as he/she went through stages of infection, stress and back to healthy. In short, our mSFPCA model generally fits this data set well, and our diagnostic tools were able to highlight biologically meaningful outliers for further examination.

Apart from modeling diversity metrics, mSFPCA can also be used to model individual omic features, such as specific taxa, cytokine, proteins and genes. Supplementary Figures 6 and 7 demonstrate the application of mSFPCA to analyze the microbial ratio between Lachnospiraceae and Bacteroides, protein IGHA2, cytokine PDGFBB and HbA1c in this data set. These specific individual omic features were selected because they were found to be the most predictive biomarkers for type 2 diabetes status (Rose et al. (2019)). Both figures show that mSFPCA can capture the mean curves and FPC curves in specific omic features well, and its performance is consistent with its results on the diversity metrics. Here, we chose to demonstrate the usage of mSFPCA on diversity metrics instead of specific features, because diversity metrics has higher prediction accuracy of type 2 diabetes status than specific features in this data set.

5. Discussion. Here, we have proposed multivariate sparse functional PCA, an extension to the sparse functional principal components analysis, to modeling multiple trajectories simultaneously. The methodological novelty lies in the computationally efficient Bayesian covariance matrix estimation, where we utilized a Cholesky decomposition to guarantee it to be positive semidefinite under the constrained form of diagonal within-trajectory covariance and arbitrary form of between-trajectory covariance structure. Moreover, we utilized mutual information to assess marginal and conditional temporal associations, providing a meaningful and interpretable measure to quantify temporal associations. Finally, our Bayesian implementation in Stan enables the usage of PSIS-LOO for efficient model selection, and visual model diagnostic methods, such as examining the estimated shape parameters from PSIS-LOO and utilizing the graphical posterior predictive checks, to evaluate the validity of mSFPCA models and highlight potential outliers.

In both our real-data based simulations and application to longitudinal microbiome multi-omics data sets, we have demonstrated that mSFPCA is able to accurately uncover the underlying principal modes of variation over time, including both the average population pattern and subject-level variation, and estimate the temporal associations properly. These enabled us to detect biologically meaningful signals in a large and challenging longitudinal cohort with irregular sampling, missing data and four data modalities. Moreover, the model diagnostics plots from real data application show that mSFPCA can provide reliable model fitting to real microbiome multiomics data set. All these results highlight the great value of our method in modeling longitudinal data with multiple data modalities. Though we applied mSFPCA to microbiome data in this paper, the method is in fact a general framework that can be applied to a wide range of multivariate longitudinal data.

The mSFPCA utilized a Cholesky decomposition to estimate the principal component covariance matrix. This approach might not be feasible for very ill-conditioned covariance matrices, that is, when two or more principal components are very close to 100% correlated.

In our experience with microbiome multiomics data, it is highly unlikely to encounter cases when Cholesky decomposition is not feasible. To demonstrate the capacity of Cholesky decomposition in modeling ill-conditioned scenarios, we simulated highly correlated trajectories and then examined the performance of mSFPCA. Supplementary Figure 4 shows that mSFPCA is able to capture the underlying mean and FPC curves well even with highly correlated principal component covariance structure, and Supplementary Figure 5 indicates that mSFPCA is still able to achieve about 95% average coverage probability in the estimation of covariance matrix. In situations where principal components are nearly perfectly correlated, users might consider alternative options (e.g., a Karhunen–Loeve decomposition) for principal component covariance estimation.

Regarding the impact of sample size, time points and missing data, we performed additional simulations to provide more information regarding the performance of mSFPCA on a multivariate data set with a different number of time points, sample size and proportion of missing data. Since we have already demonstrated the robust performance of SFPCA to limited sample size (as low as 10 subjects) and a high proportion of missing data (as high as 80%) with extensive simulations in earlier work (Jiang et al., 2020), with regards to the estimations of mean and FPCs, we choose to highlight the performance of mSFPCA in covariance estimation here, which is unique to mSFPCA. We expanded the earlier mSFPCA simulations in the simulation section by varying the total number of subjects at 50, 100, 200, the total number of time points at 5, 10, 20, and the proportion of missing values at 20%, 50% and 80%. Results over all 100 simulated data sets for each scenario were summarized in Supplementary Table 1, which shows clearly that mSFPCA is able to achieve desired 95% coverage probabilities despite the changes in the number of sample size, the total number of time points and the proportions of missing data. It would be challenging to precisely define the lowest number of time points, minimum sample size and lowest threshold for missing data required in mSFPCA method, because it depends on a number of other factors as well, such as the temporal spacing of the assessments across subjects, and the dynamic temporal change of the measurements over time. Based on our simulation results, mSFPCA is able to handle data with 50 subjects, the number of time points as low as 5 and the proportion of missing data as high as 80%. But the best way to check whether mSFPCA is appropriate for a specific data set is to use our graphical diagnostic tools to examine whether the model fits the data well.

Although theoretically there is no limitation on the number of temporal measurements mSFPCA can model simultaneously, computational cost does grow exponentially with the increasing number of temporal measurements, sample size and data modalities. Supplementary Figure 8 shows the average computation time across 100 replicates for simulated data sets (with the number of sample size varying at 50, 100, 500, the total number of time points at 5, 10 and 20 and the proportion of missing data at 20%), by using 3 Markov chains with 1000 iterations. The results from Supplementary Figure 8 indicate that computation time increases when the number of subjects increases or when the total number of time points increases. On our processor (Intel(R) Xeon(R) Processor 2.4 GHz, 125 GiB memory), it takes about 26 minutes for mSFPCA on a simulated data set with 3 data modalities, 100 subjects, 10 time points and 20% missing data. The computational time increases by about 1.5 fold when the data modalities increase to 5 (72 minutes), and requires an additional 2 fold of time when the data modalities increase to 10 (240 minutes). Thus mSFPCA can be practically applied to longitudinal multivariate data with a fairly high number of data modalities, but there are practical limitations due to the computational performance of our implementation of the Bayesian MCMC algorithm. Increased computational efficiency might be achieved by migrating our code from R and Stan into Julia, a new programming language for data and analytics that combines the functionality of quantitative environments such as R and Python

with the speed of production programming languages like Java and C++. In particular, Julia provides parallel and distributed computing capabilities, and unlimited scalability.

Another limitation of our method is that we assume the principal component scores and residuals to be normally distributed as in the original SFPCA model. This normality assumption would restrict our method from being applying to highly skewed trajectories. However, improper application of the method to such data could be detected by the model diagnostic tools we provide, for example, the graphical posterior predictive model checks. Users could also modify the mSFPCA model by incorporating alternative prior distributions, for example, a t-distribution with a low degree of freedom to capture heavy tails in the distribution of principal component scores, which can be easily implemented in Stan. Finally, since the mSFPCA model is implemented in Stan, a programming language with a very active user base, this method will be able to be updated with more efficient MCMC sampling algorithms and also incorporate other groundbreaking model selection and diagnostic techniques whenever they become available. Hence, we believe that the mSFPCA method will become a useful and up-to-date tool for researchers in various fields to analyze longitudinal data with multiple measurements in order to detect complex temporal associations.

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SUPPLEMENTARY MATERIAL

Code and data (DOI: [10.1214/21-AOAS1587SUPPA](https://doi.org/10.1214/21-AOAS1587SUPPA); .zip). To ensure reproducibility, the data set used in this manuscript, and the code for simulations and real data analysis are provided in this zip file.

Supplement to “Multivariate sparse functional principal components analysis for longitudinal microbiome multiomics data” (DOI: [10.1214/21-AOAS1587SUPPB](https://doi.org/10.1214/21-AOAS1587SUPPB); .pdf). Detailed description of constrained Cholesky decomposition for covariance estimation, five supplementary figures for simulation results, two supplementary figures for application results and one supplementary table and one supplementary figure regarding the discussion section are included in the supplementary materials.

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